COSC 325 Midterm Report

Paul Adams
The University of Tennessee Knoxville
padams20@vols.utk.edu

Carlo Duran
The University of Tennessee Knoxville
cduran3@vols.utk.edu

Andreas Papas
The University of Tennessee Knoxville
apapas@vols.utk.edu

Carter Holtmann
The University of Tennessee Knoxville
cholman@vols.utk.edu

I. Introduction

This model aims to help instructors and proofreaders identify the author of a text as human or artificial intelligence. With the advent of easy access to large language models, the amount of AI-generated text has exploded. Furthermore, as the machine learning and artificial intelligence industries continue to grow, it has become increasingly difficult to tell human- and AI-written text apart. Being able to provide a reliable razor to determine the origin of an excerpt, would give editors and instructors more confidence that the work they receive reflects the opinion or argument of a human author rather than a crowd-sourced answer from the internet. This could help identify cases of plagiarism or cheating and eventually reduce it altogether by discouraging would-be offenders. The goal of this project is two-fold. First is to identify consistent differences between human- and AI-written text. Second, is the development of a machine learning model that can reliably label text as either written by a person or a large language model. A novel and very interesting technique was explored in a recent study published on 5 October 2026 that was conducted by Rajendhran et al. [1]. Their work investigates not only the difference in word choice made by AI and humans, but more importantly how they express ideas differently and form distinct logical arguments and progressions of thought. The key element that distinguishes the model they developed, IdeaLens, is that it attempts to identify the source of the ideas conveyed by the writing. Articles like the one by Hellström in 2024 have shown that human writing has gradually become more like that of AI [2]. As such, the detection of AI-written work has become much more difficult when combined with the rapid improvement of AI as a whole. Therefore, this new approach may become necessary to reliably distinguish between the words of human and machine.

II. Dataset

This project's dataset includes excerpts of various kinds from both human and AI authors. Each piece of data has the following associated features:

- text content
- content type
- word count
- character count
- sentence count
- lexical diversity
- average sentence length
- avg word length
- punctuation ratio
- flesch reading ease
- grammar errors
- passive voice ratio
- predictability score
- burstiness
- sentiment score
- label

Text-content contains the excerpt itself, while content-type describes the excerpt's subject matter. The features word-count, character-count, and sentence-count denote the number of words, characters, and sentences that make up each excerpt respectively. Punctuation-ratio, on the other hand, quantifies the proportion of each excerpt which punctuation constitutes. Flesch-reading-ease and gunning-fog-index are different measures of how easy or difficult it is to read a passage. Grammar-errors is a count of the number of errors relating to grammar that are made in each excerpt. Passive-voice-ratio concerns itself which the proportion of the passage that is in the passive voice. Predictability-score measures how easy or difficult it is to guess each following word in a passage. Burstiness is a way of quantifying the variety in sentence length, structure, and writing style in each passage. The sentiment-score is a numerical score assigned to each excerpt based on the emotional tone and pacing it conveys. The label marks an excerpt as either human- or machine-written.

Due to the correlation between the word-count, character-count, and sentence-count features, we decided to remove

word-count and sentence-count. Additionally, the flesch-reading-ease was stripped from the data to further reduce feature collinearity. Furthermore, we used median interpolation to fill in any missing numeric values in the data. Following the feature selection, all the data were normalized according to a min-max distribution. When it comes to general trends within the data, a few seemingly unintuitive ones cropped up such as an inverse proportionality between the punctuation-ratio and the avg-sentence length, but these are quite logically explained upon further examination. For instance, the aforementioned correlation between punctuation-ratio and avg-sentence-length is more easily understood when considering that periods are punctuation and the average length of each excerpt is similar. However, the AI confidence scores in figure 2 indicate that the model seems to perform best when given more than two but less than ten samples.

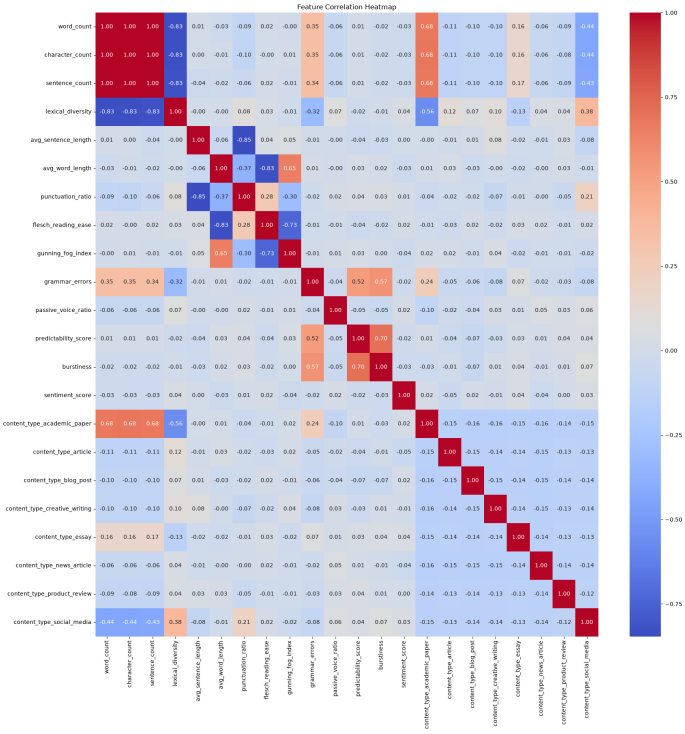


Fig. 1. Fig 1. Feature Correlation heatmap. Note the high dimensionality of the data.

IV. Proposed Improvements

During the remainder of the project, we plan to significantly improve our model by incorporating additional sub-models and improving the accuracy of the logistic regression classifier. An ensemble model would likely provide an edge over already-existing AI detection models, so we plan to incorporate a decision-tree based model in addition to our regression classifier. We would need to write a more comprehensive program to determine which questions would result in the most information, as the richness of our data results in a situation where it may not be immediately clear which decision split we should start with at the root of the decision tree. Furthermore,

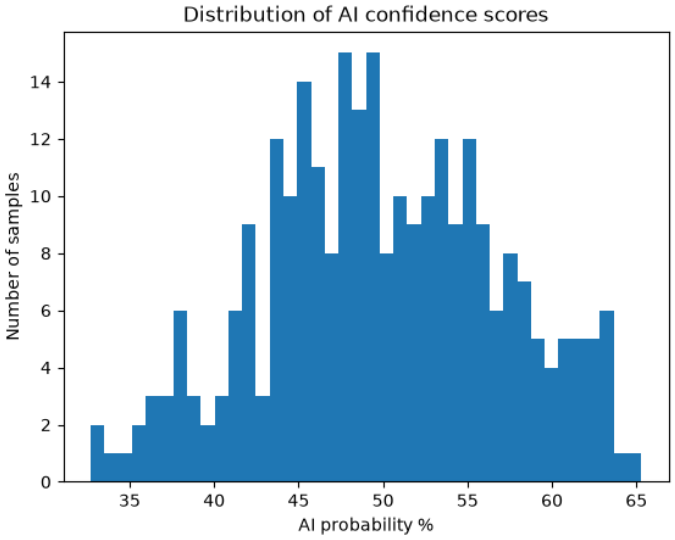


Fig. 2. Fig 1. Feature Correlation heatmap. Note the high dimensionality of the data.

III. Baseline Model

Our baseline model was a Sklearn logistic regression classifier. After cleaning the data and removing correlated features, we had hoped that a simple regression would be applicable to the problem. Owing to the high dimensionality of our data, we decided to use a logistic regression classifier. However, our model did not perform well initially; at the moment, it has an average of 0.46 accuracy. Since our accuracy was below 0.50, it could also be that our model is better at making incorrect assumptions, and by simply inverting the prediction outputs we may be able to achieve 0.54 accuracy. This accuracy is still not impressive, but it does belie a greater confusion–why our model is failing to establish correlations at all. Since our accuracy is near 0.50, it's as though the model is making blind guesses, which indicates that it has failed to determine which features actually correlate with AI-written work and which correlate with human-generated text. We need to investigate whether our process of feature selection chose features which had little to do with determining the nature of a text, and which feature selection processes we could use to improve our selection. Our model's prediction range was also slightly concerning. As seen in Figure 2, our model's predictions followed a roughly normal distribution with a median at 0.46, but the predictions are skewed slightly toward AI-detected works, indicating that our model tended to think more works were AI-written than not, when the dataset was, presumably, equally split. This discrepancy could be caused by an incorrect emphasis on poorly-correlating features, which would lead to the model assuming certain features indicated AI-created work when they did not. In addition, the confusion matrix for our model shows roughly similar counts in each category, which further implies that our model is making random guesses. As before, this issue could be solved by paying better attention to our feature selection.

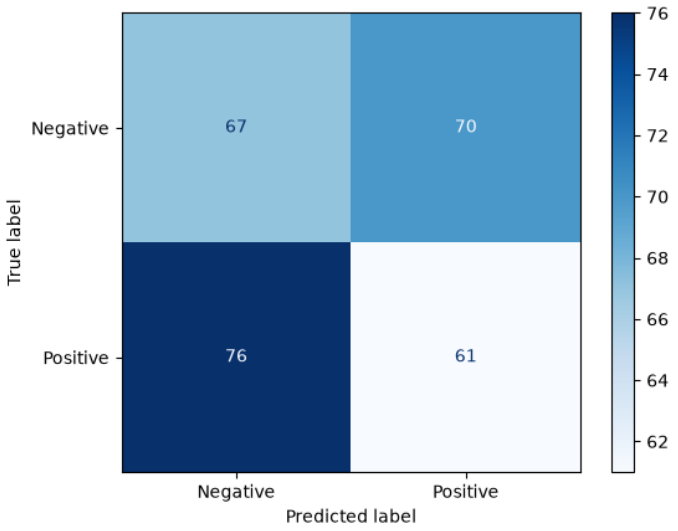


Fig. 3. Fig 1. Feature Correlation heatmap. Note the high dimensionality of the data.

this tree would need a large number of splits, and will be computationally expensive. Regardless, if the tree is able to produce results at greater than 0.46 accuracy, it will have easily been worth it. Alternatively, or perhaps additionally, we may quickly train a linear regression model to determine whether linear or logistic regression is the better solution. While our data has a high number of dimensions–lending itself to logistic regression–our output predictions are formatted as a range from 0 to 1 representing the probability of text being AI-generated, which lends itself well to linear regression techniques. Our finished ensemble model will primarily be evaluated based on prediction accuracy, as that appears to be our biggest downfall at the moment. To this end, we can improve our model accuracy by determining which features are most important. SKlearn, by default, uses gradient descent to determine feature weights. This approach should be a good fit for our data, as we have a high number of dimensions; however, we may try different approaches, such as Sequential Backward Selection or Permutation Feature Importance. Sequential Backward selection would help reduce the dimensionality of our data, which would be helpful when we implement the decision-tree framework. Permutation Feature Importance would also help parse the high dimensionality of the dataset. Random Forest Feature Selection was also considered, but owing to the high dimensionality of our data, we determined it would not be an effective method.

V. Work Distribution

During the first half of this project, work was distributed on an equal-effort basis. Andreas and Carter were responsible for the data cleaning, model production, and data visualization. To this end, Andreas wrote the code for primary data-cleanup and processing; Carter wrote the code for the logistic regression classifier and plotted the results, and both worked together to generate the data visualization graphs present in this report. Paul and Carlo were responsible for the literature review and compositing of this report; Carlo wrote the first half of the report, and Paul wrote the second half of the report, including this paragraph.

References

[1] R. Rajendhran et al., "IdeaLens: Detecting AI Ideas in Long-Form Writing," Oct. 05, 2026. Accessed: Oct. 21, 2026. [Online]. Available: https://arxiv.org/abs/2610.06778v1

[2] T. Hellström, "AI and Its Consequences for the Written Word," Frontiers in Artificial Intelligence, vol. 6, Jan. 2024, doi: 10.3389/frai.2023.1326166.